

[Outline](#) > 2. Improving the model > 2.2 Practice problem (Flu!) > Calculation

Calculation

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The differential equation for the fraction of people $X(t)$ on Ledom Island infected with the flu virus is

$$\frac{dX}{dt} = k X(t) (1 - X(t)),$$

where k is a positive parameter. The initial condition is $X(0) = 0.02$.

Exercise A

1/1 point (graded)

The equilibrium solutions of the differential equation are

☒ $X(t) = 0$ ✓☐ $X(t) = 0.02$ ☒ $X(t) = 1$ ✓☐ $X(t) = 1000$ [Submit](#)

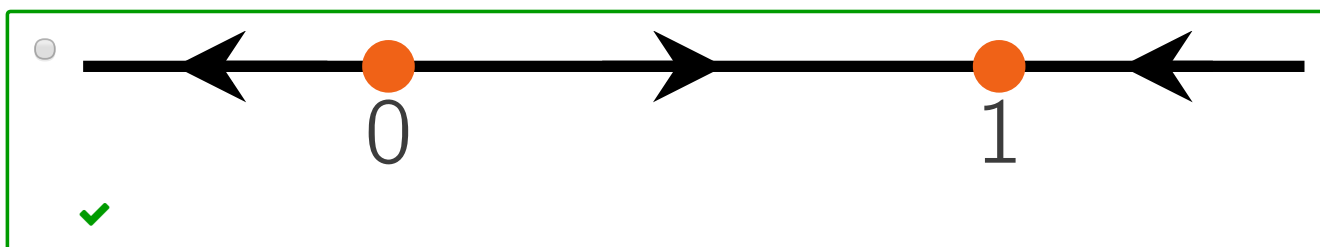
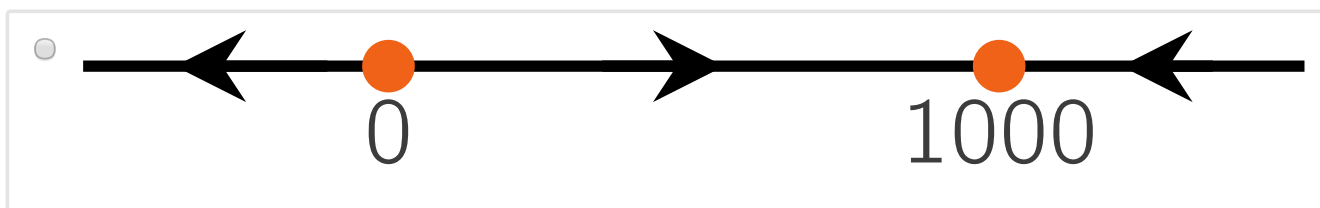
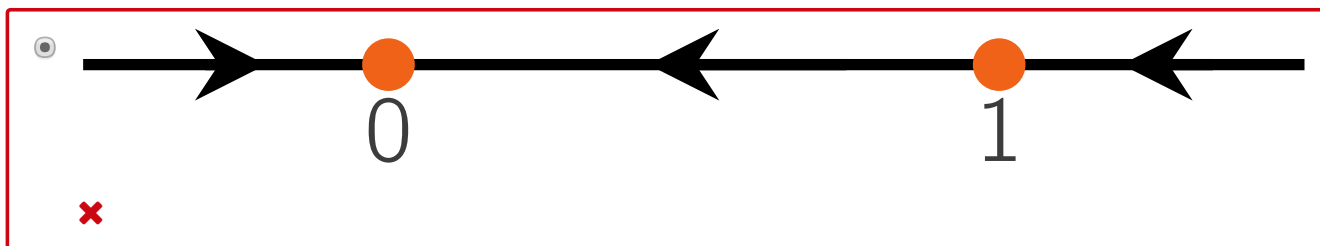
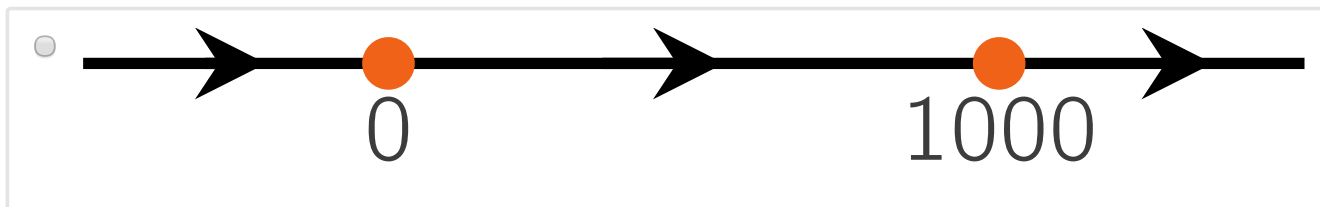
You have used 1 of 2 attempts

i Answers are displayed within the problem

Exercise B

0/1 point (graded)

The phase line for the differential equation is



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You have used 1 of 1 attempt

i Answers are displayed within the problem

Exercise C

1/1 point (graded)

The smallest equilibrium solution is

☐ stable

☒ unstable ✓

The largest equilibrium solution is

☒ stable ✓

☐ unstable

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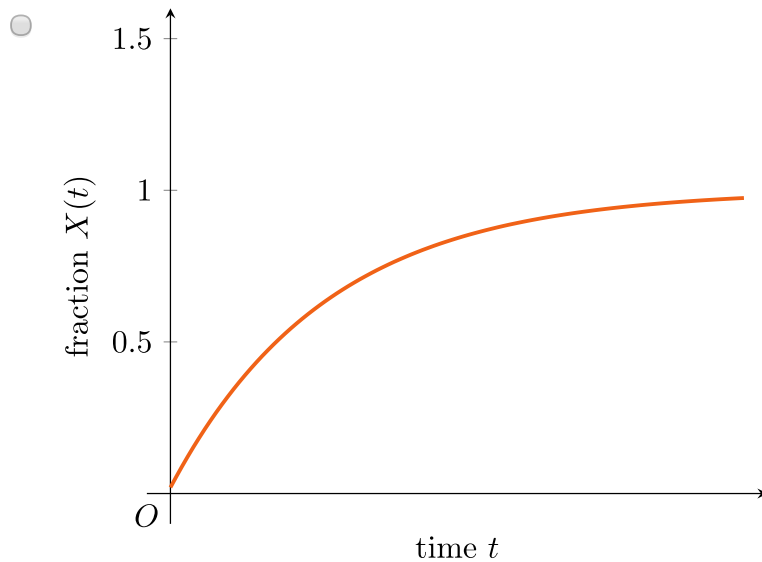
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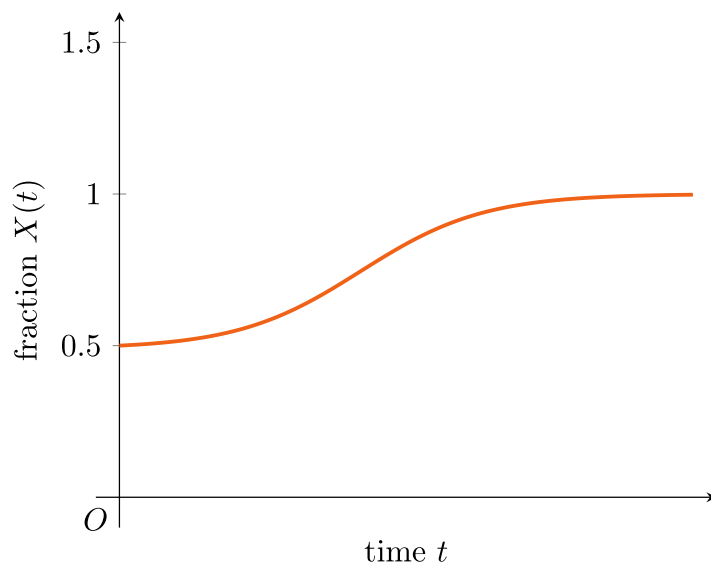
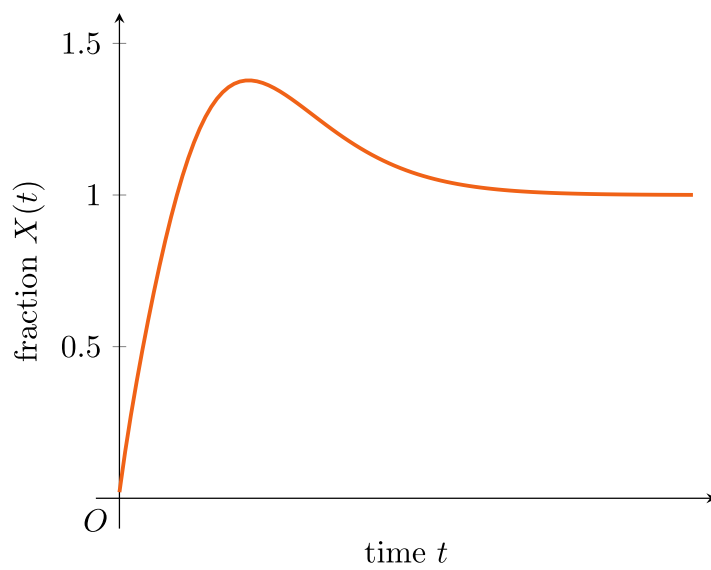
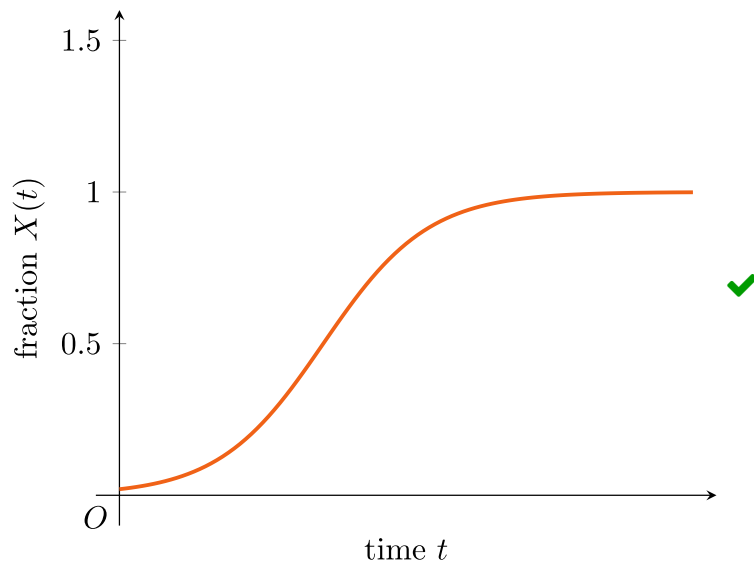
i Answers are displayed within the problem

Exercise D

1/1 point (graded)

The graph of the solution $X(t)$ is





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You have used 1 of 1 attempt

i Answers are displayed within the problem

On the next page the validation.

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